Ingeniería de Software

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Entrega 04 - Diagramas UML

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Diagrama de Clases1

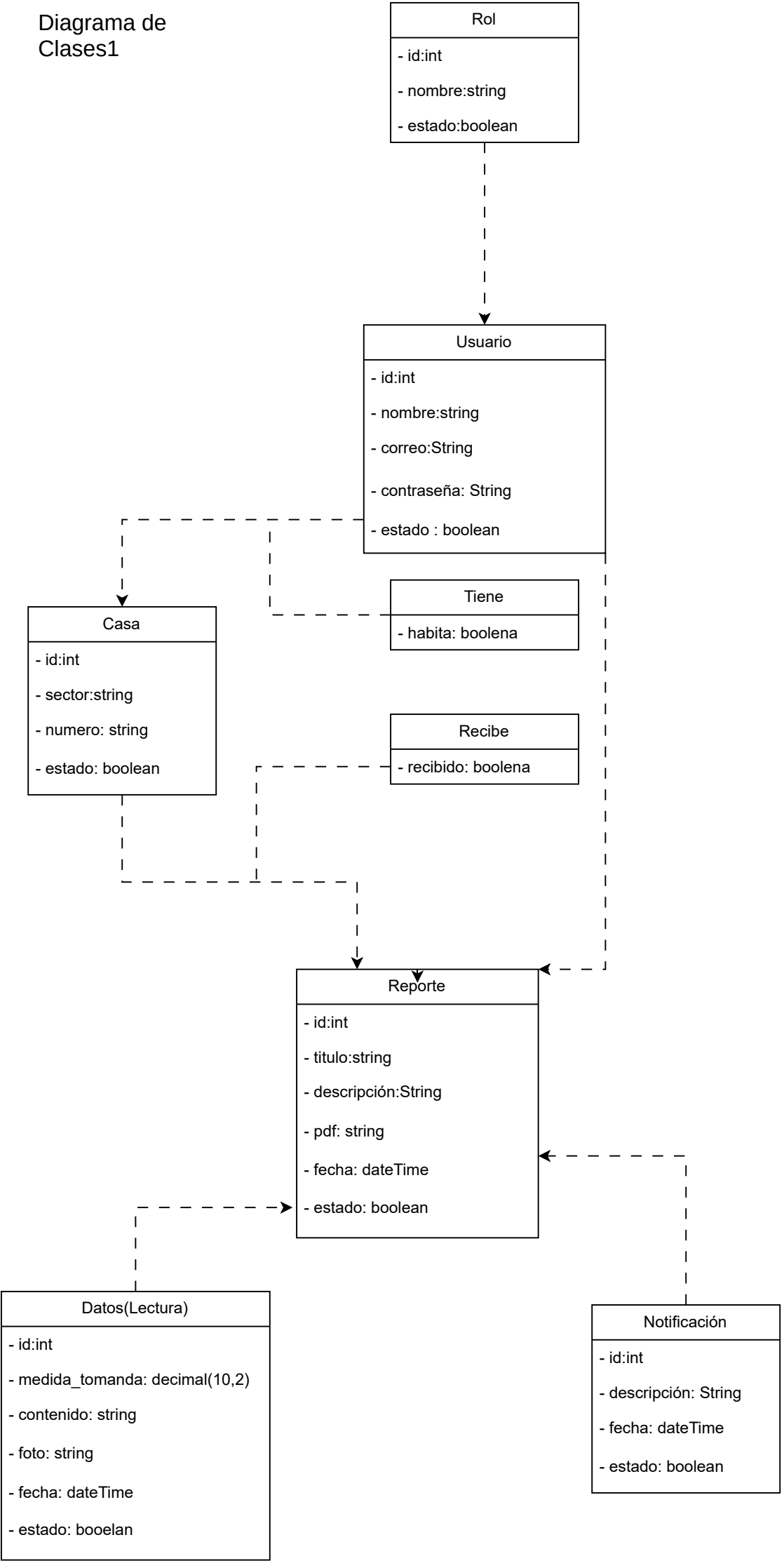


Diagram Heading

Diagrama de Clases 2

StateLessWidget

StatefulWidget
+ createState() : State

State
+ init()
+ dispose()
+setState()
+build(context): Widget

MyApp
+build(context):widget

LoginScreen

_LoginScreenState
-eyeopen:bool
-cargando:bool
-correoController: TextEditingController
-contrasenaController: TextEditingController
+iniciarSesion:Future<void>

AdminScreenRouter

UserScreenRouter

MstaffScreenRouter

UsersReportsScreen

_UserReportsScreenState
-casas : List<Map>
-reportes : List<Map>
-casaSeleccionada : int
-mesSeleccionado : DateTime
+obtenerReportes() : Future<void>
+reportesFiltrados() : List<Map>
+marcarRecibido(reporteId : int) : Future<void>
+descargarReporte(reporte : Map)

_MstaffScreenRouterState
-indexNav : int

MstaffReportScreen

MstaffInputScreen
+casaInicial:Map

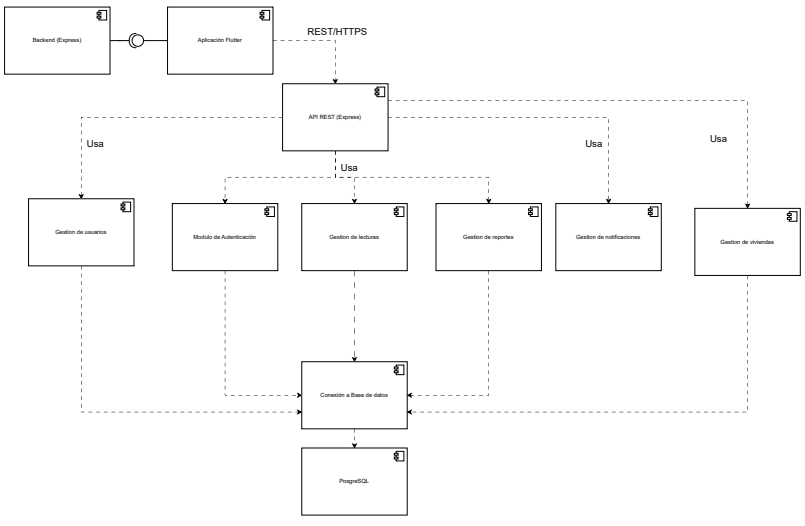
_MstaffReportsScreenState
-reportes : List<Map>
-reporteProcesando : int
+obtenerReportes() : Future<void>
+emitirReporte(id : int) : Future<void>

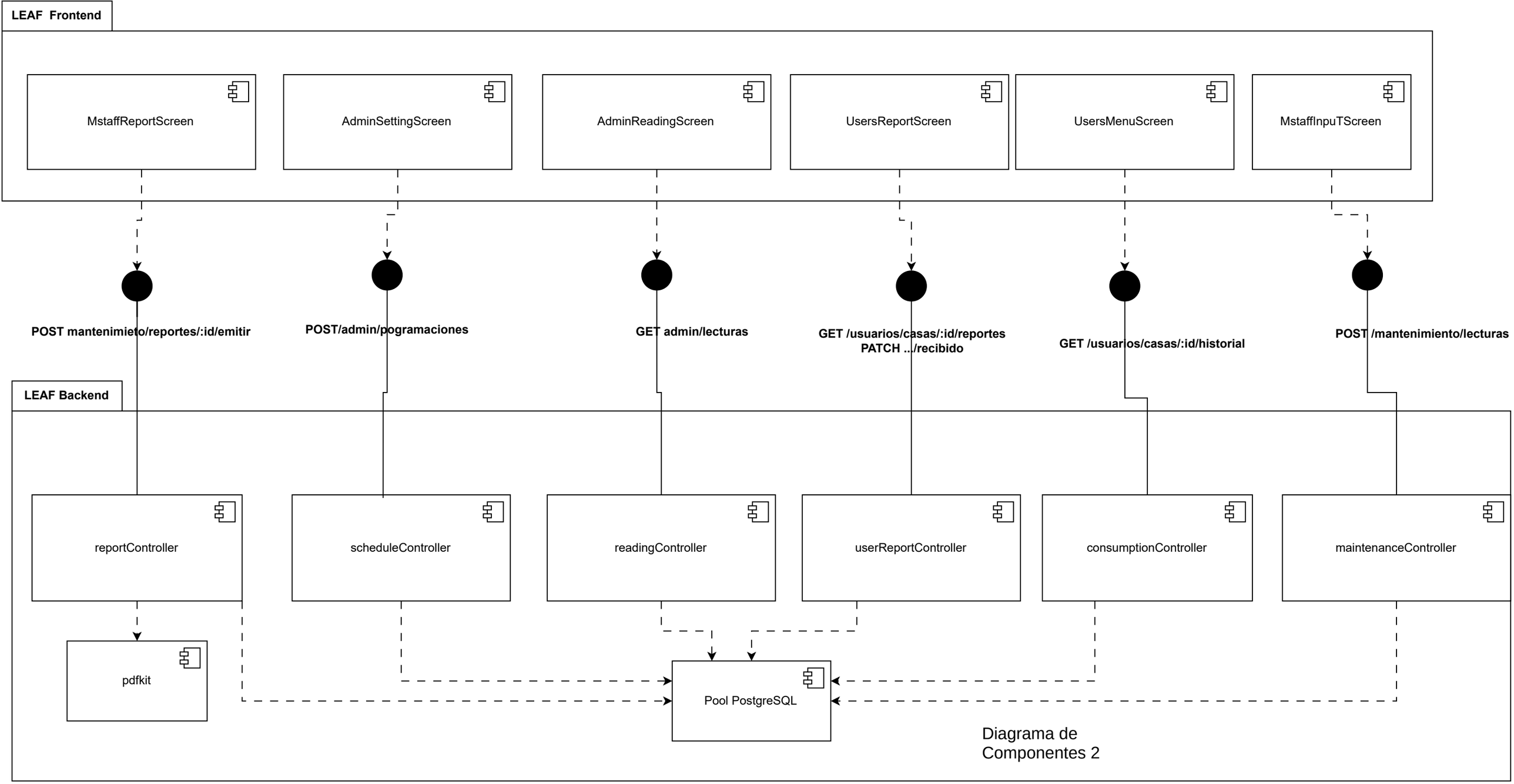
_MstaffInputScreenState
-casas : List<Map>
-casaSeleccionada : int [0..1]
-cargando : bool
-guardando : bool
-medidaController : TextEditingController
-contenidoController : TextEditingController
+obtenerCasas() : Future<void>
+guardarLectura() : Future<void>
+mostrarMensaje(texto, color)

APIservice
+instance : ApiService
+baseUrl : String = "http://localhost:3000"
+storage : FlutterSecureStorage
+token : String
+usuario : Map
-ApiService._()
+iniciarSesion(correo, contrasena) : Future<void>
+recuperarSesion() : Future<void>
+cerrarSesion() : Future<void>
+get(endpoint) : Future<Map>
+post(endpoint, body) : Future<Map>
+put(endpoint, body) : Future<Map>
+patch(endpoint, body) : Future<Map>
+procesarRespuesta(response) : Map

AppColors
+lightBlue : Color
+meidumBlue : Color

Diagrama de Componentes 1




LEAF Frontend
MstaffReportScreen
AdminSettingScreen
AdminReadingScreen
UsersReportScreen
UsersMenuScreen
MstaffInpuTScreen
POST mantenimieto/reportes/:id/emitir
POST/admin/pogramaciones
GET admin/lecturas
GET /usuarios/casas/:id/reportes
PATCH .../recibido
GET /usuarios/casas/:id/historial
POST /mantenimiento/lecturas
LEAF Backend
reportController
scheduleController
readingController
userReportController
consumptionController
maintenanceController
pdfkit
Pool PostgreSQL
Diagrama de
Componentes 2

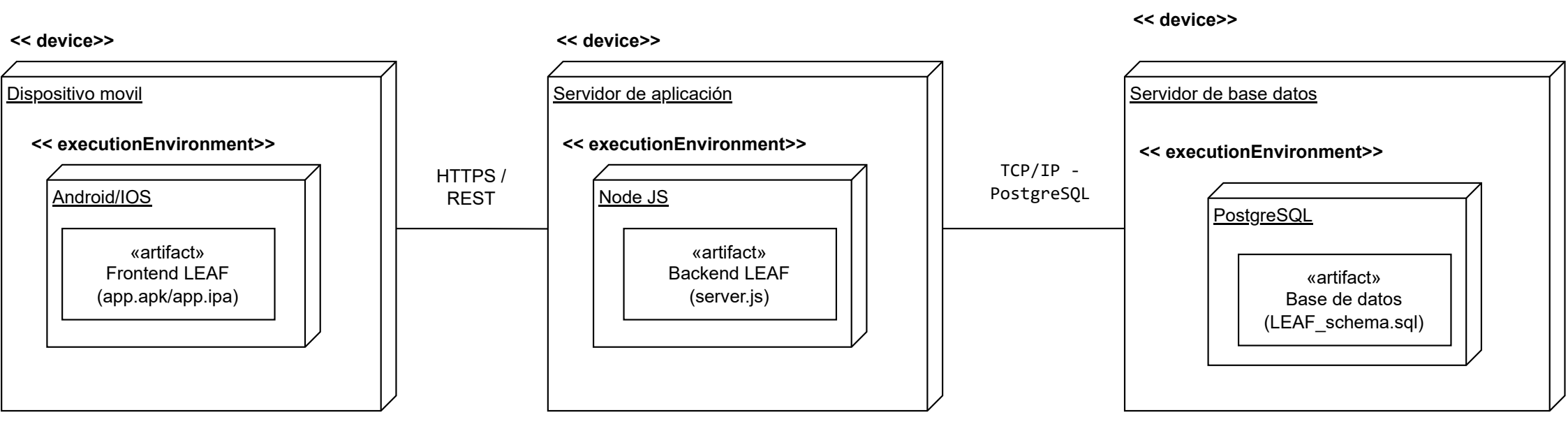
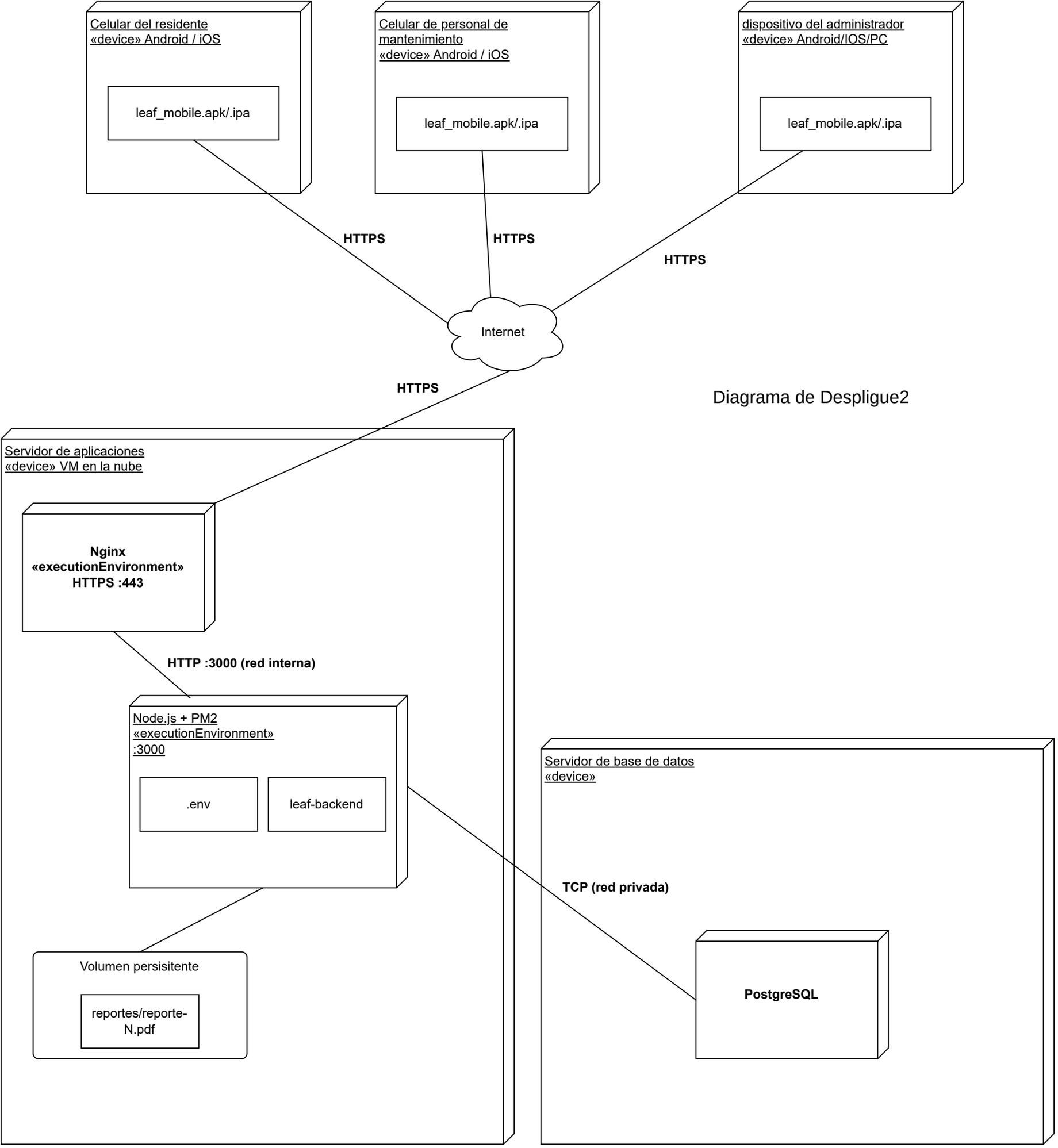


Diagrama de Despligue1


Celular del residente
«device» Android / iOS
leaf_mobile.apk/.ipa
Celular de personal de mantenimiento
«device» Android / iOS
leaf_mobile.apk/.ipa
dispositivo del administrador
«device» Android/IOS/PC
leaf_mobile.apk/.ipa
HTTPS
HTTPS
HTTPS
Internet
HTTPS
Diagrama de Despligue2
Servidor de aplicaciones
«device» VM en la nube
Nginx
«executionEnvironment»
HTTPS :443
HTTP :3000 (red interna)
Node.js + PM2
«executionEnvironment»
:3000
.env
leaf-backend
Volumen persisitente
reportes/reporte-N.pdf
Servidor de base de datos
«device»
TCP (red privada)
PostgreSQL

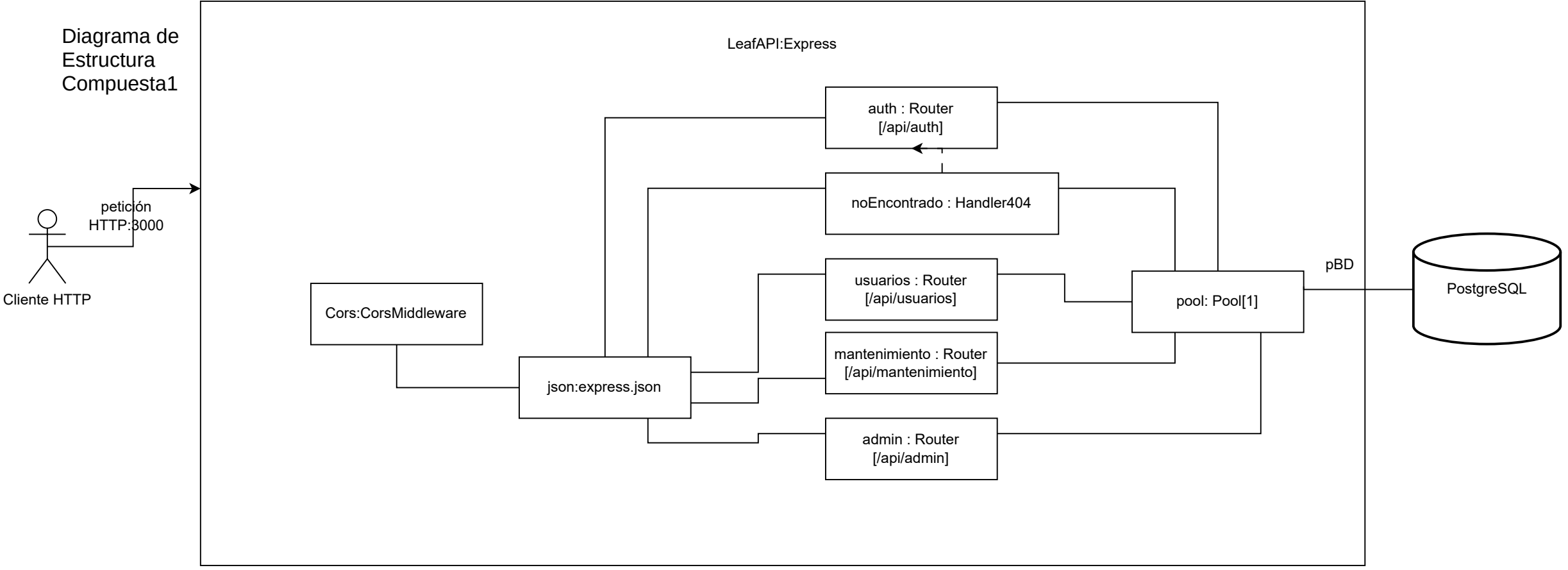
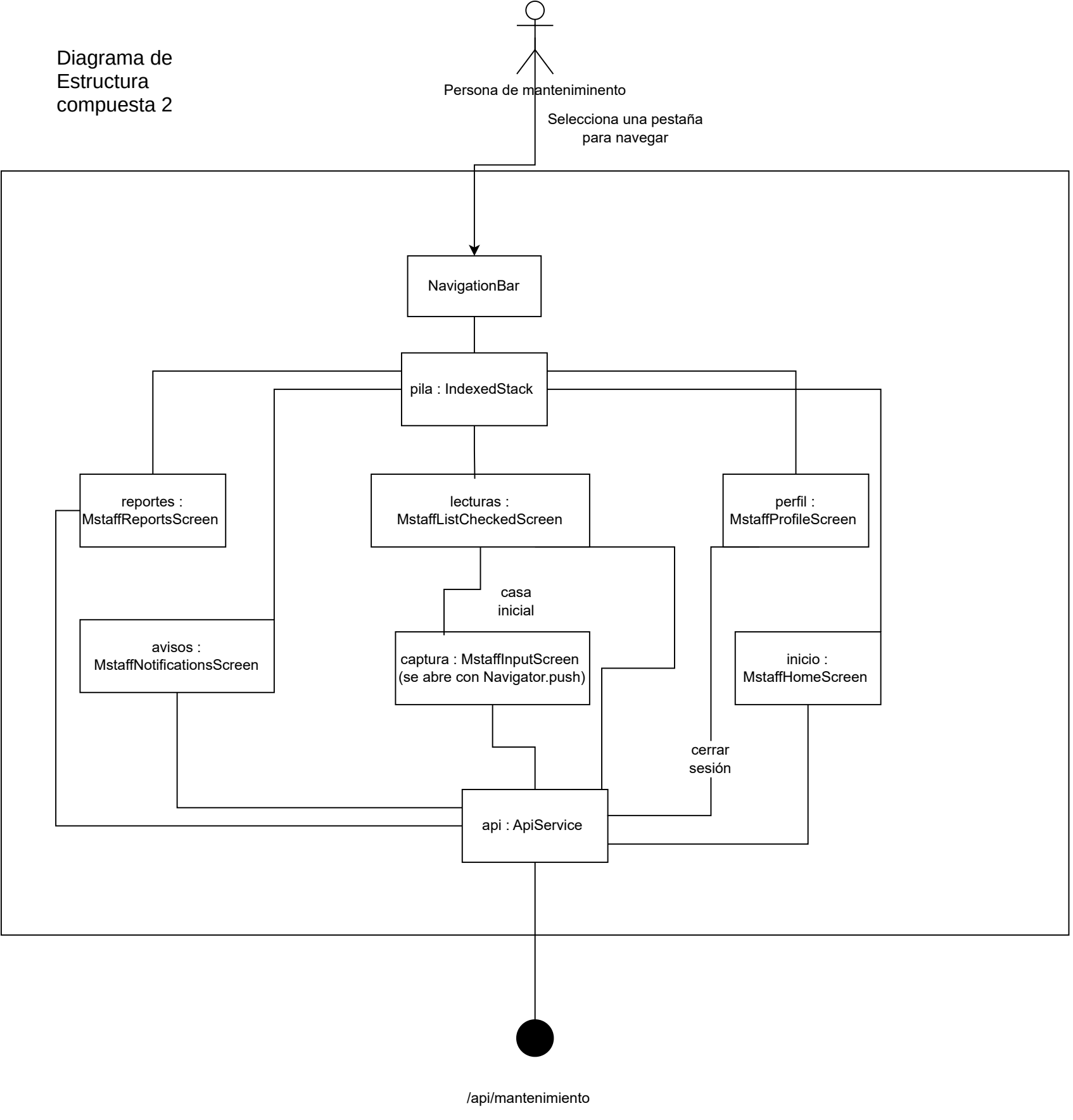
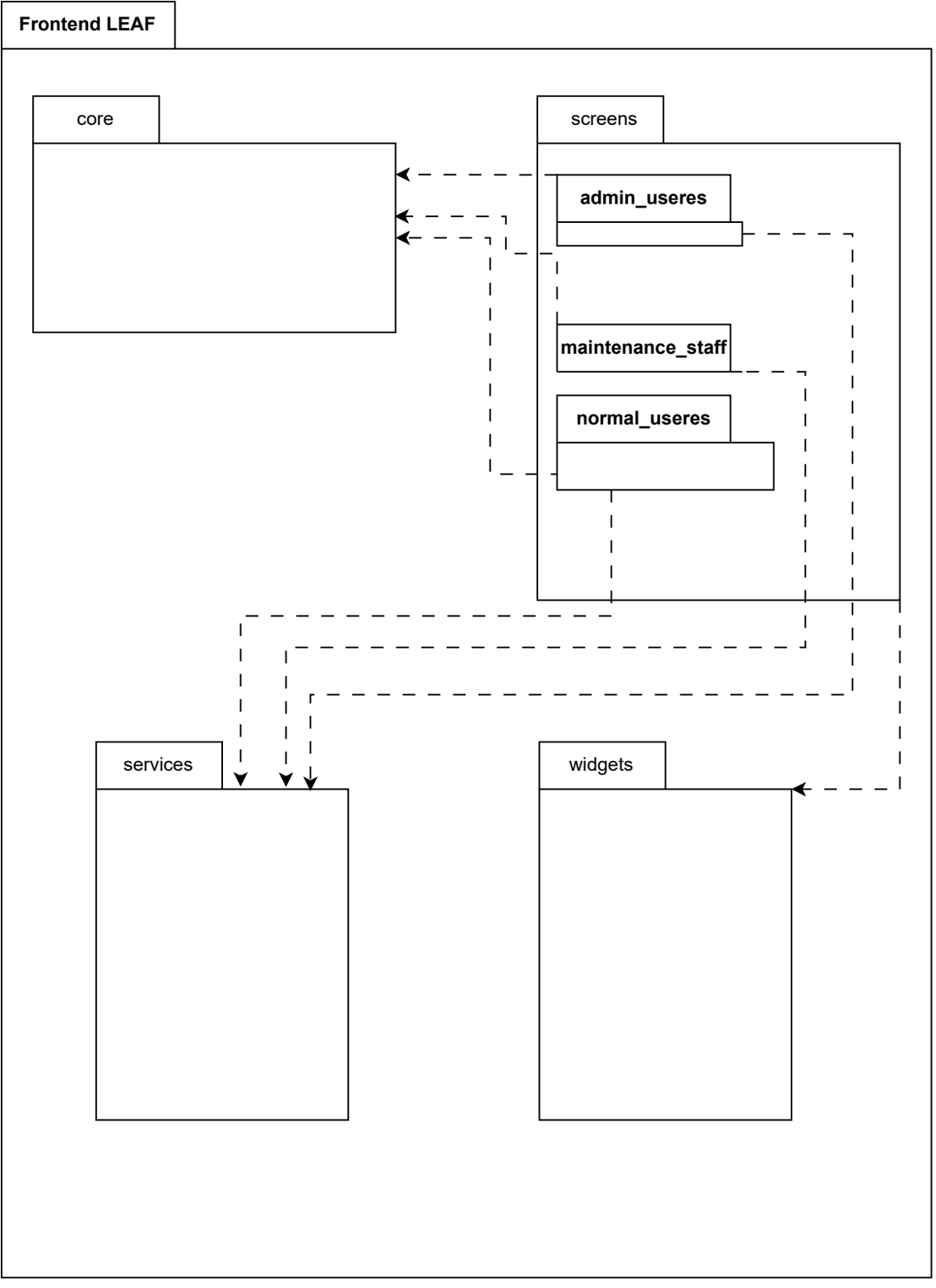

Diagrama de Estructura Compuesta1
LeafAPI:Express
auth : Router
[/api/auth]
noEncontrado : Handler404
usuarios : Router
[/api/usuarios]
mantenimiento : Router
[/api/mantenimiento]
admin : Router
[/api/admin]
Cors:CorsMiddleware
json:express.json
pool: Pool[1]
pBD
PostgreSQL
petición
HTTP:3000
Cliente HTTP

Diagrama de Estructura compuesta 2





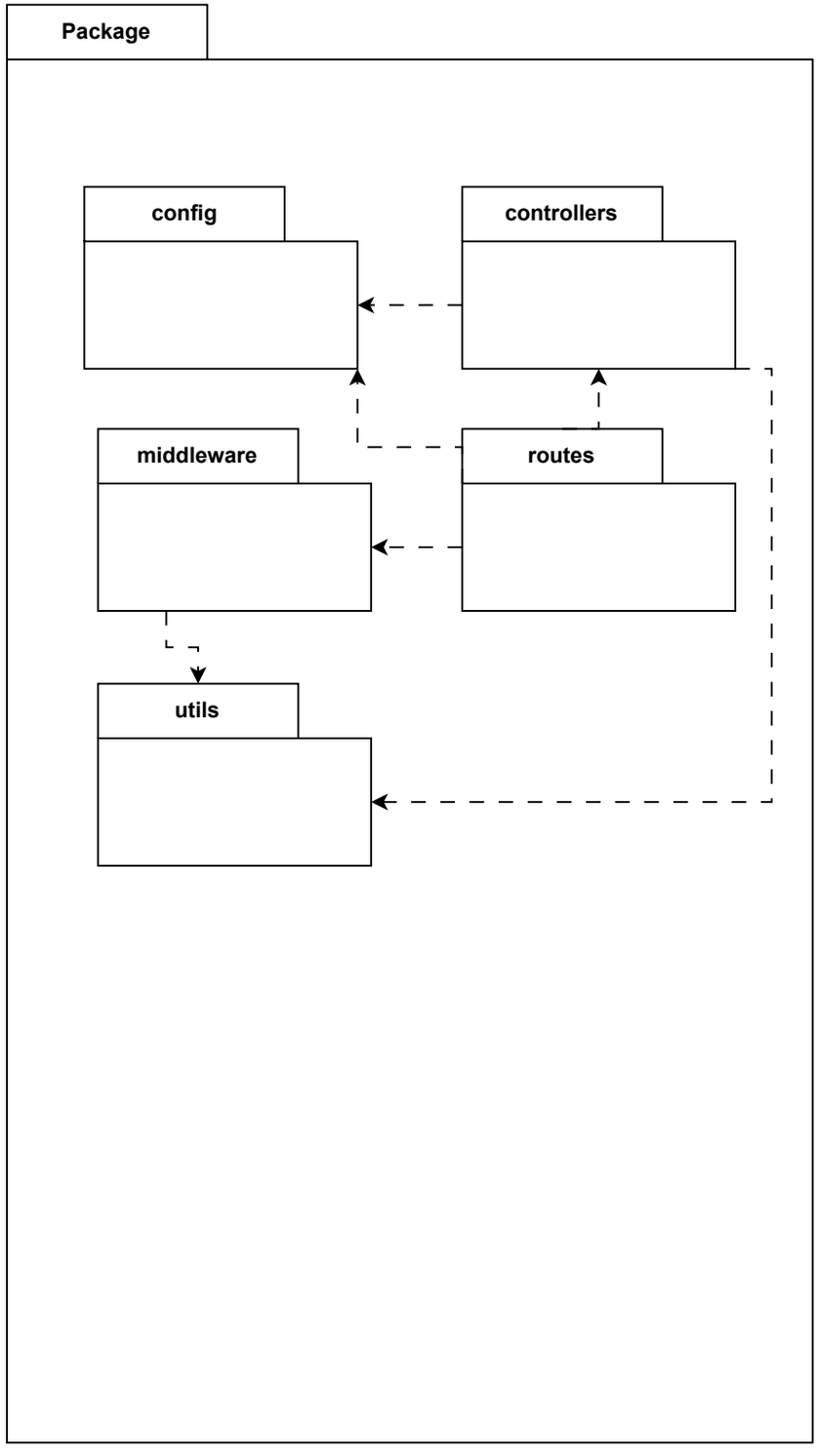
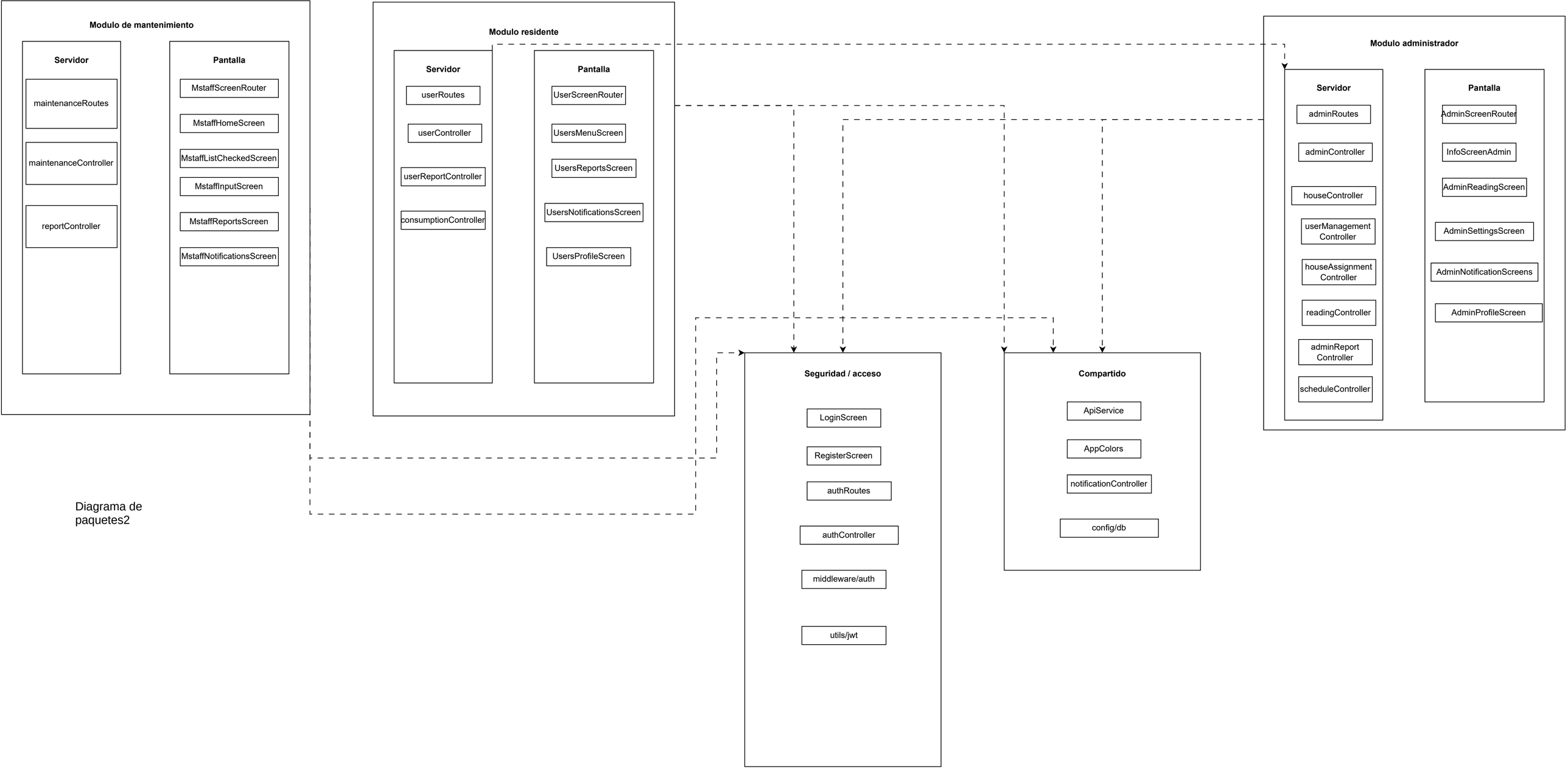


Diagrama de Paquetes1


Modulo de mantenimiento
Servidor
maintenanceRoutes
maintenanceController
reportController
Pantalla
MstaffScreenRouter
MstaffHomeScreen
MstaffListCheckedScreen
MstaffInputScreen
MstaffReportsScreen
MstaffNotificationsScreen
Modulo residente
Servidor
userRoutes
userController
userReportController
consumptionController
Pantalla
UserScreenRouter
UsersMenuScreen
UsersReportsScreen
UsersNotificationsScreen
UsersProfileScreen
Modulo administrador
Servidor
adminRoutes
adminController
houseController
userManagement Controller
houseAssignment Controller
readingController
adminReport Controller
scheduleController
Pantalla
AdminScreenRouter
InfoScreenAdmin
AdminReadingScreen
AdminSettingsScreen
AdminNotificationScreens
AdminProfileScreen
Seguridad / acceso
LoginScreen
RegisterScreen
authRoutes
authController
middleware/auth
utils/jwt
Compartido
ApiService
AppColors
notificationController
config/db
Diagrama de paquetes2

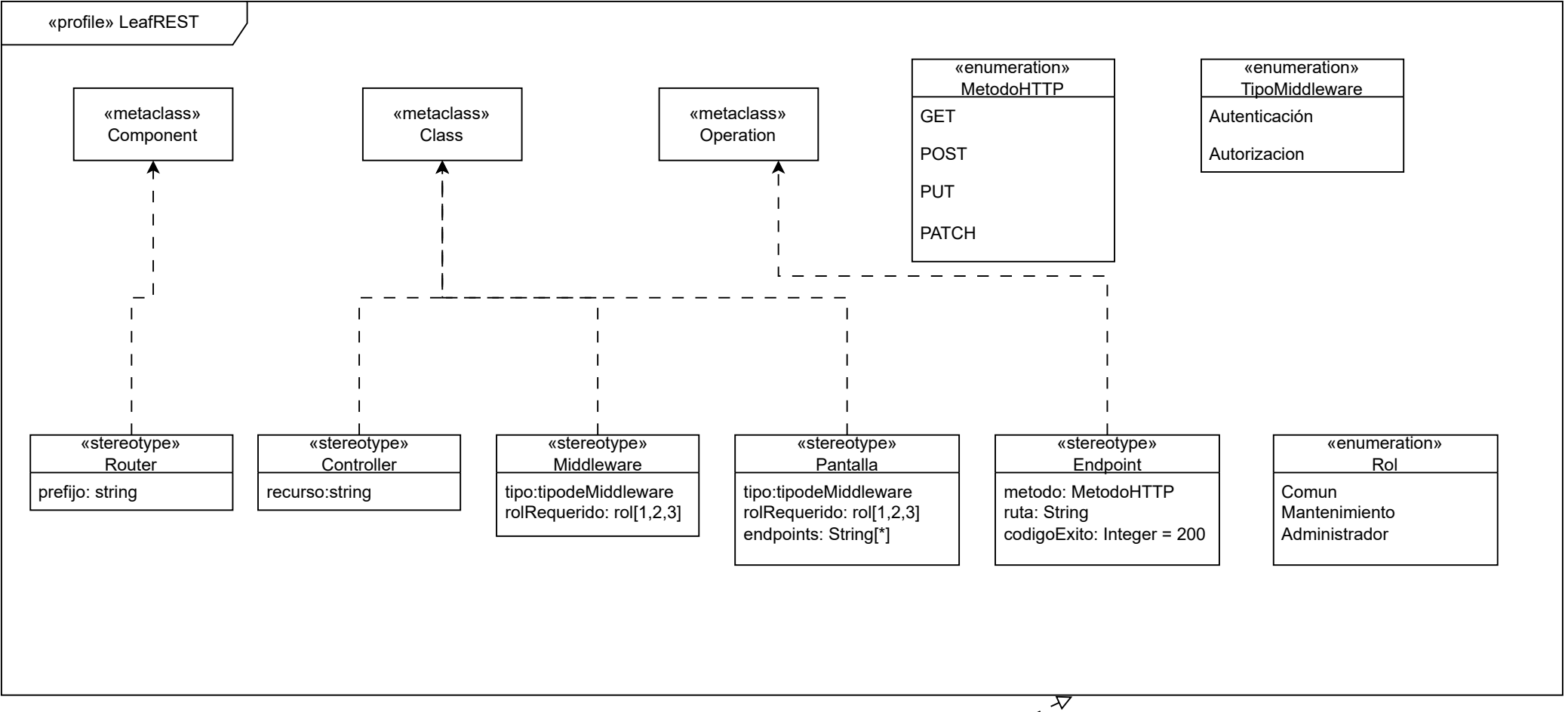
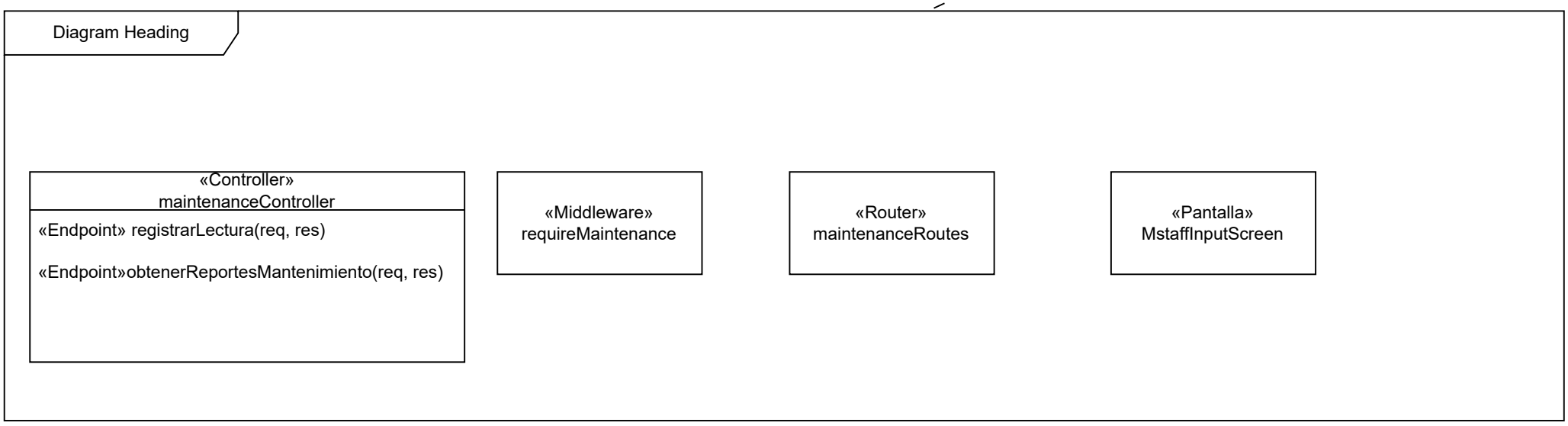
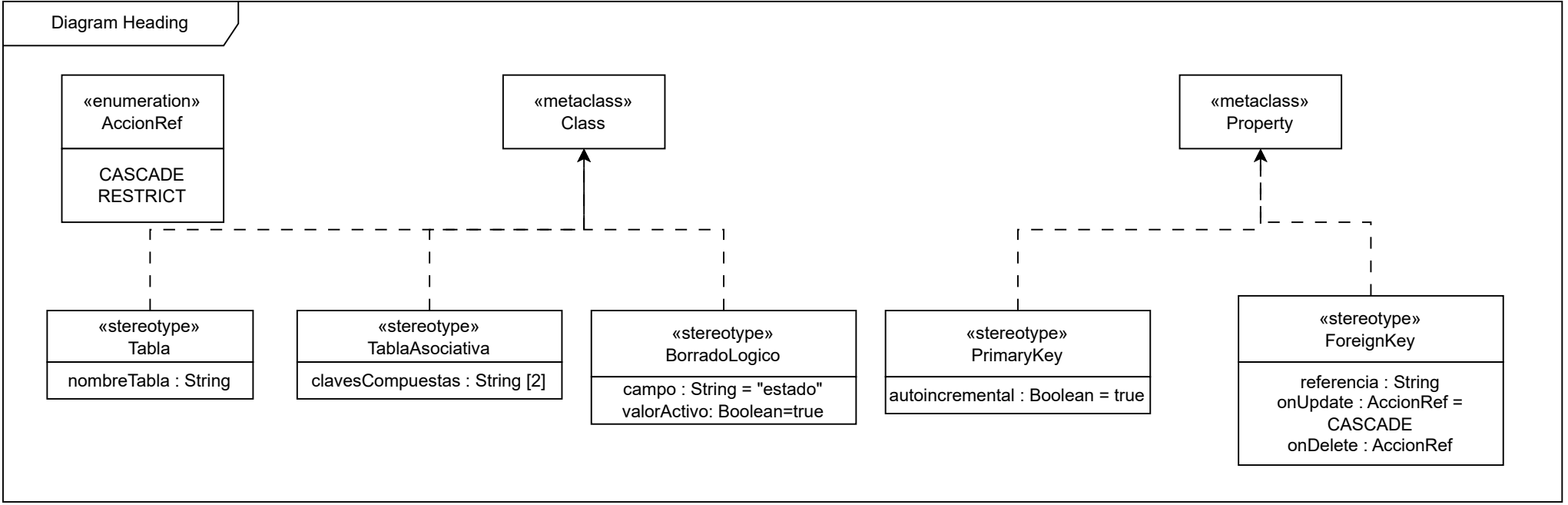


Diagrama de Perfiles1





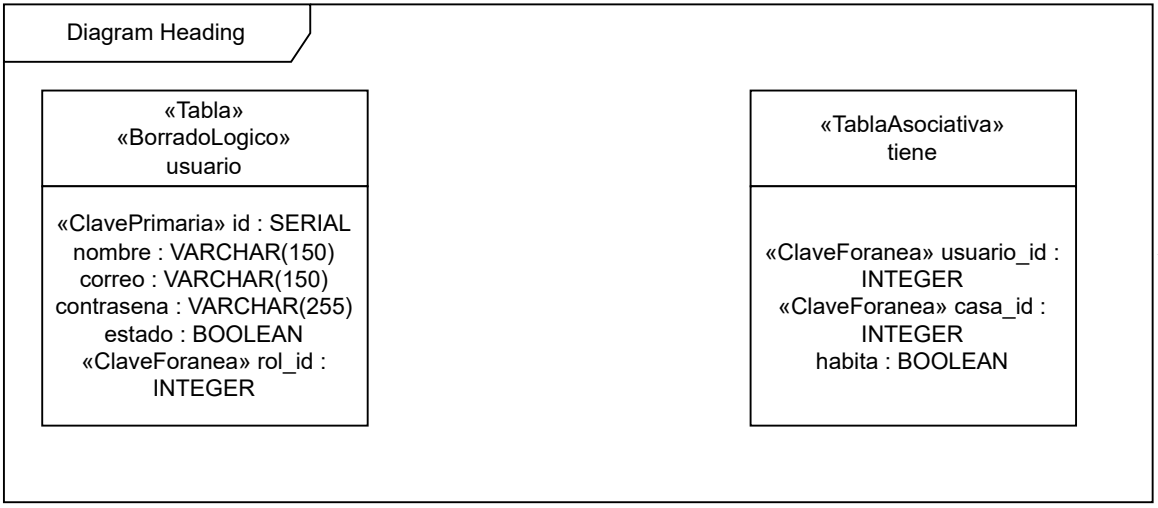
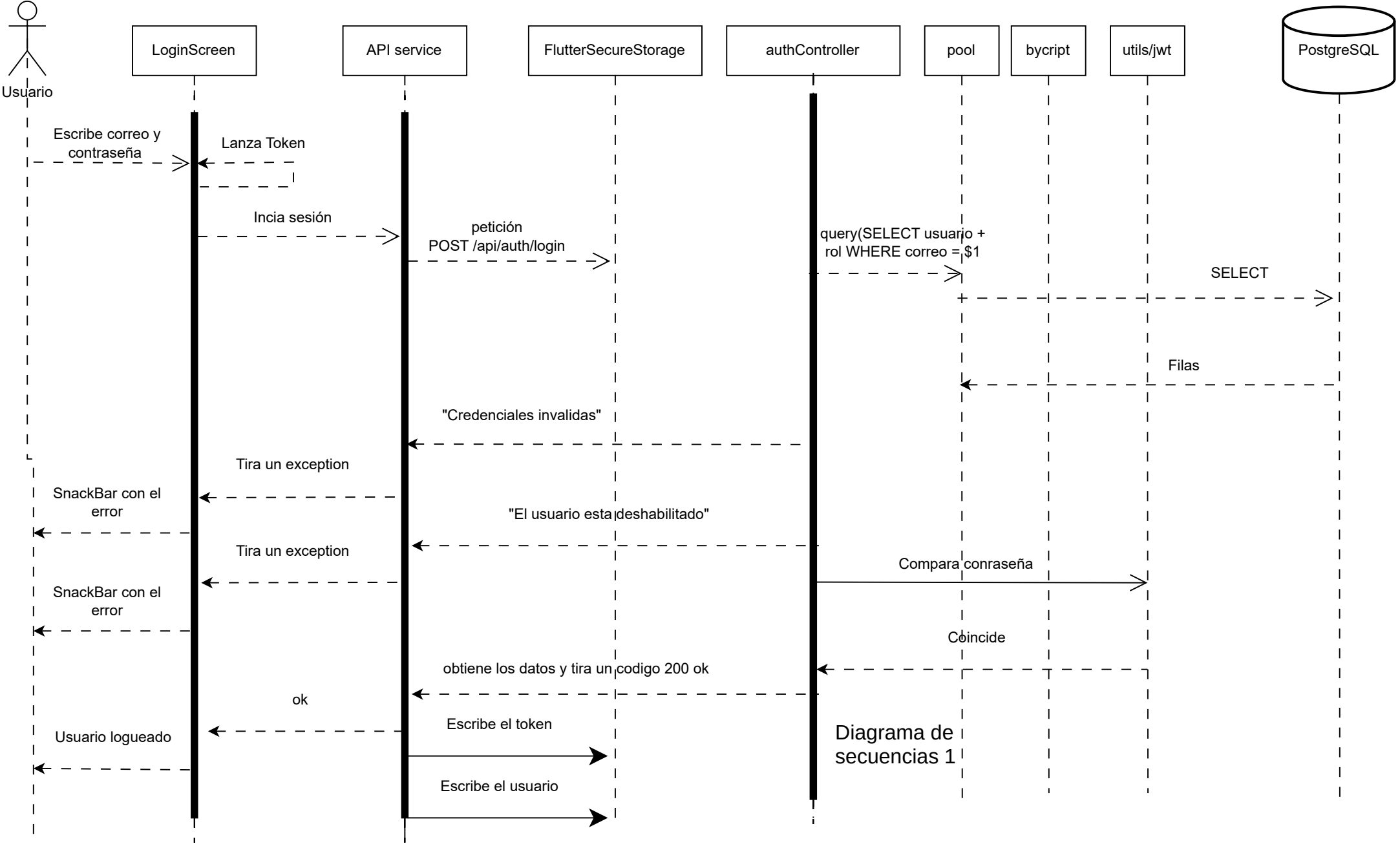


Diagrama de Perfiles2


Usuario
LoginScreen
API service
FlutterSecureStorage
authController
pool
bycript
utils/jwt
PostgreSQL
Escribe correo y contraseña
Lanza Token
Incia sesión
petición
POST /api/auth/login
query(SELECT usuario + rol WHERE correo = $1
SELECT
Filas
"Credenciales invalidas"
Tira un exception
SnackBar con el error
"El usuario esta deshabilitado"
Tira un exception
Compara conraseña
SnackBar con el error
Coincide
obtiene los datos y tira un codigo 200 ok
ok
Escribe el token
Usuario logueado
Escribe el usuario
Diagrama de secuencias 1

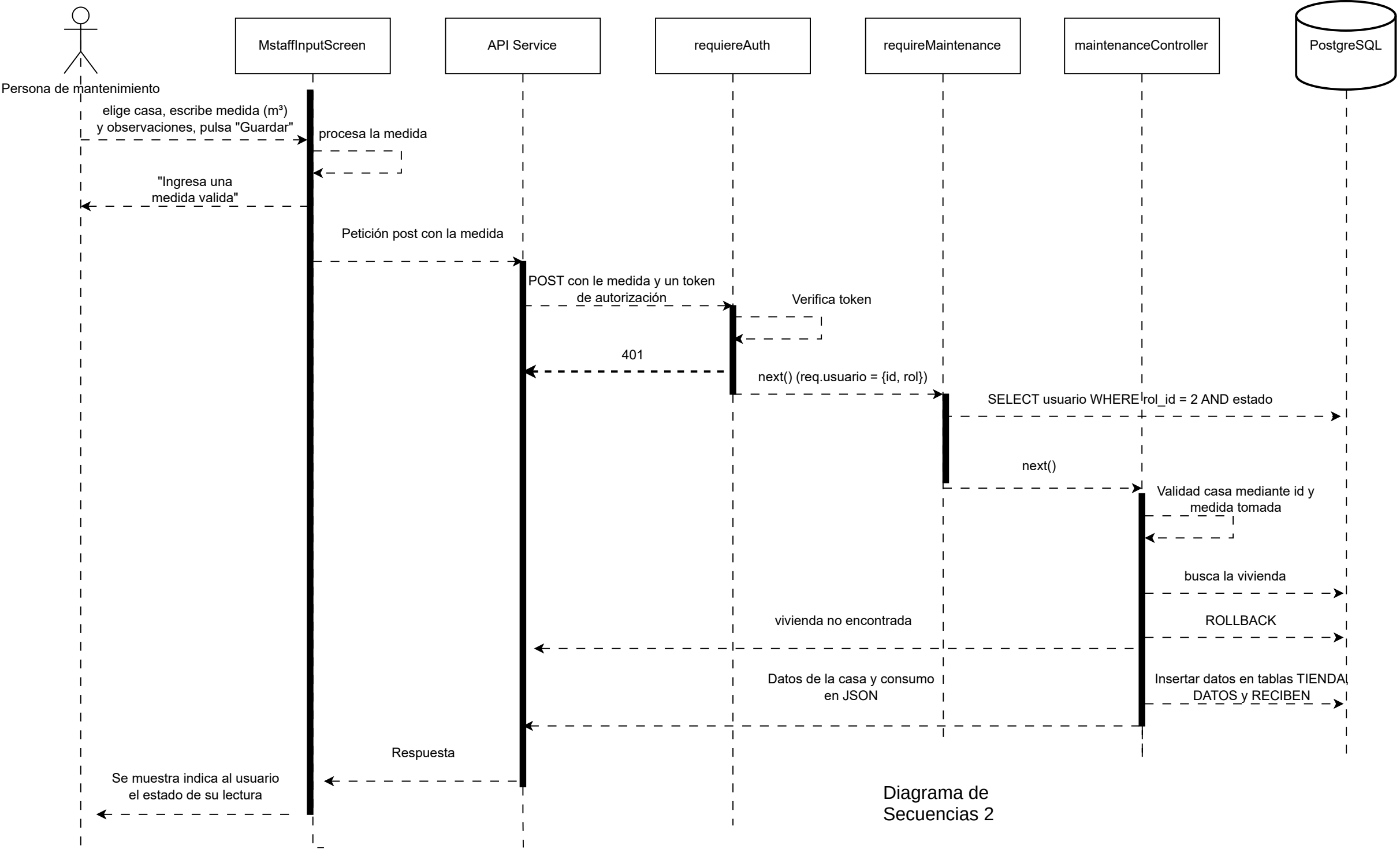

Persona de mantenimiento
MstaffInputScreen
API Service
requiereAuth
requireMaintenance
maintenanceController
PostgreSQL
elige casa, escribe medida (m³)
y observaciones, pulsa "Guardar"
procesa la medida
"Ingresa una
medida valida"
Petición post con la medida
POST con le medida y un token
de autorización
Verifica token
401
next() (req.usuario = {id, rol})
SELECT usuario WHERE rol_id = 2 AND estado
next()
Validad casa mediante id y
medida tomada
busca la vivienda
vivienda no encontrada
ROLLBACK
Datos de la casa y consumo
en JSON
Insertar datos en tablas TIENDA,
DATOS y RECIBEN
Respuesta
Se muestra indica al usuario
el estado de su lectura
Diagrama de
Secuencias 2

Diagrama Caso de Uso

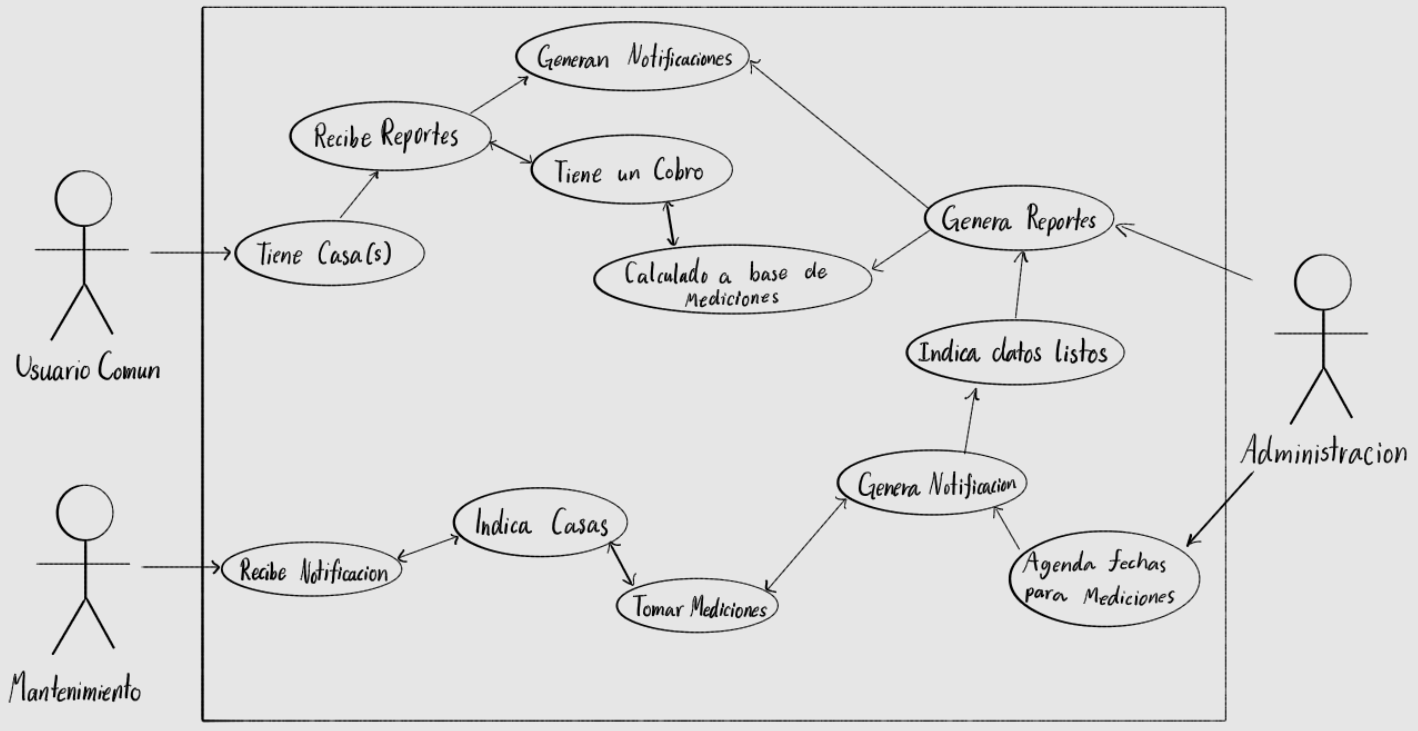


Diagrama de Actividades

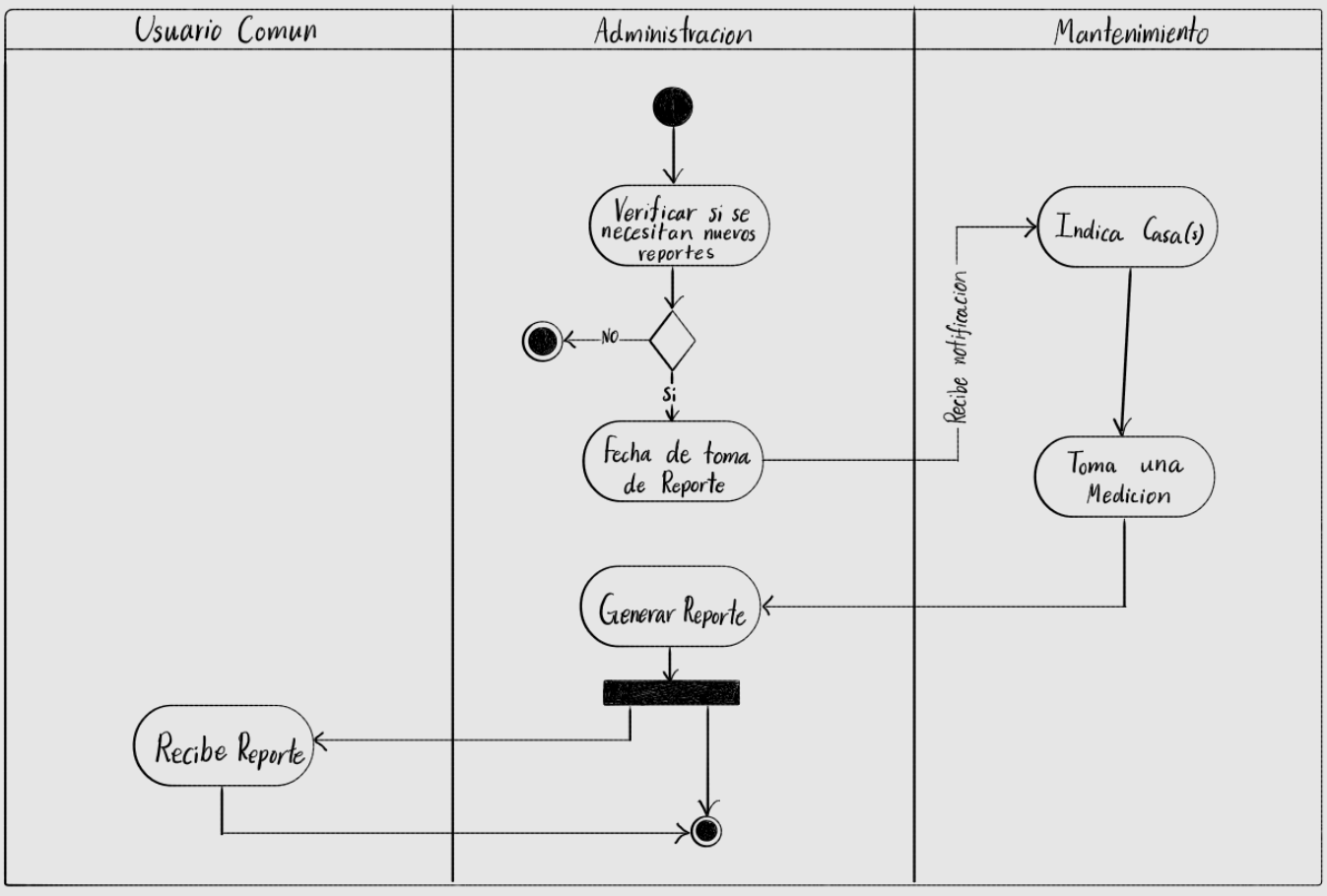


Diagrama de Estados

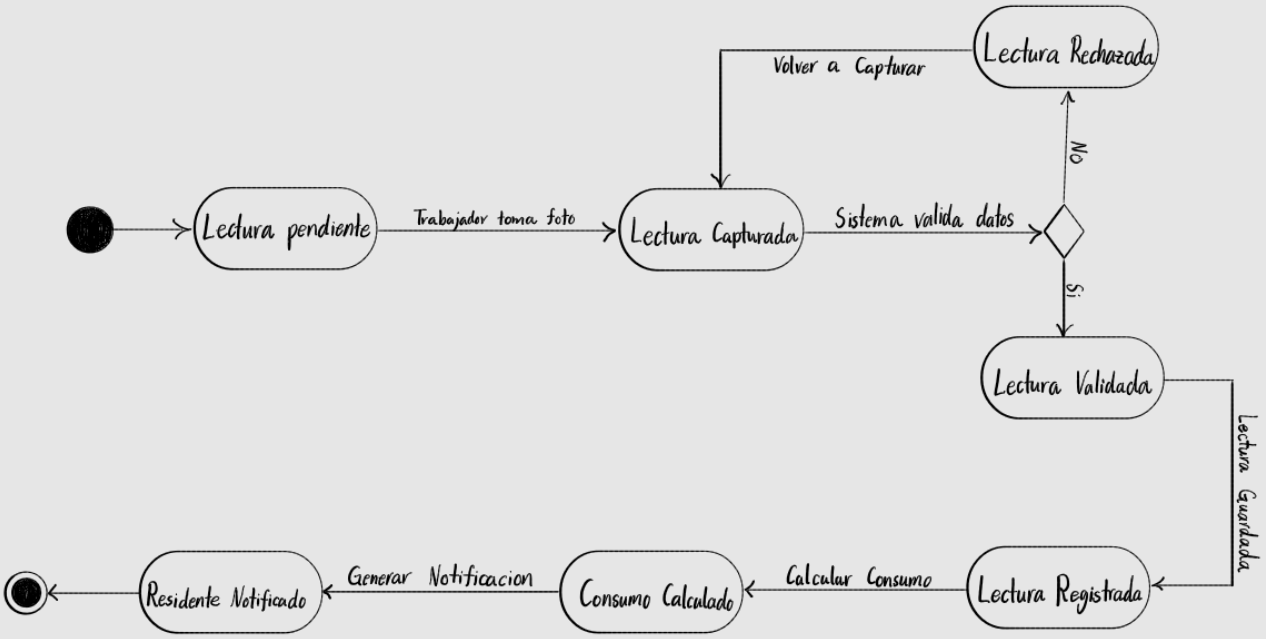


Diagrama de Objetos

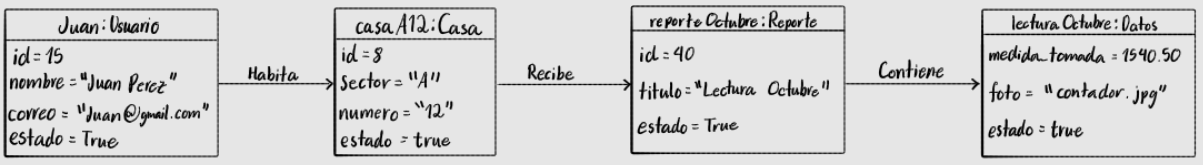


Diagrama de Comunicacion

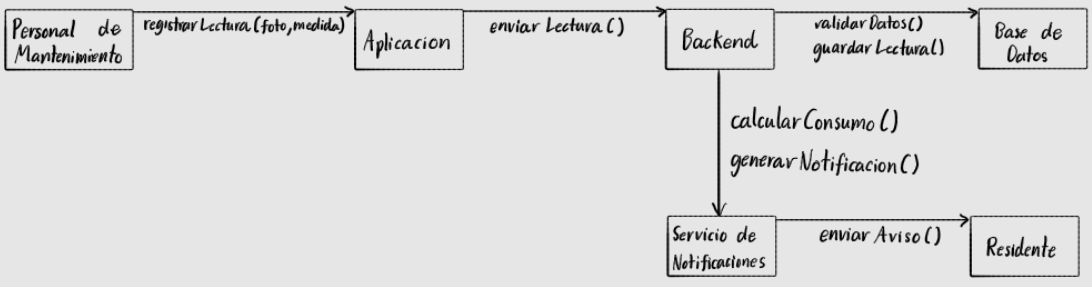


Diagrama de Tiempo

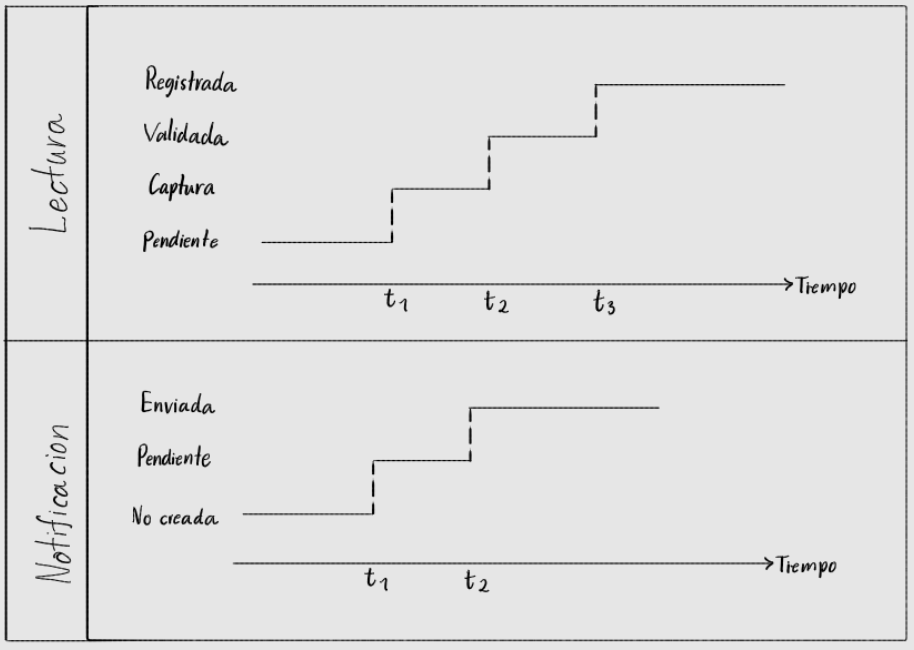


Diagrama de Interacciones

